FUNDAMENTALS OF DATA ANALYTICS

ACADEMIC

Grade 12

Curriculum Guide

FUNDAMENTALS OF DATA ANALYTICS

Pre-requisite: None

Course Description:

This course introduces foundational principles and practices in data analytics, while developing essential 21st-century skills. Learners enhance their information, visual, media, technology, and digital literacy through data collection, cleaning, processing, visualization, storytelling, and the responsible use of digital tools. The course also fosters creativity, critical thinking, problem solving, teamwork, collaboration, and communication skills in performing data analytics for informed decision making in data-driven fields. Throughout the course, learners gradually develop a project that applies data analytics concepts to solve real-world problems using datasets prepared for presentation by the end of the course.

Elective: Academic
Time Allotment: 80 hours for one term

CONTENT	**CONTENT STANDARDS** *The learners demonstrate an understanding that...*	**LEARNING COMPETENCIES** *The learners...*
	Data Analytics	
1. Data	1. data analytics involves understanding different types of data, the stages of the data life cycle, and the various types of data analysis that support informed and evidence-based decisions.	1. define data analytics. 2. differentiate data types. 3. discuss data life cycle and types of data analysis.
2. Data Privacy and Ethics	2. data privacy and ethical principles guide the responsible collection, handling, and analysis of data, ensuring that individuals' rights are protected and that	4. explain the importance of data privacy. 5. discuss data ethics in performing tasks related to data analytics.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding that...*	LEARNING COMPETENCIES *The learners...*
	analytics practices adhere to legal and moral standards.	
3. Applications of Data Analytics in Real-Life,	3. data analytics is widely applied across real-world contexts and industries, and that the field requires specific skills and offers diverse career opportunities.	6. identify real-world applications of data analytics. 7. identify skills that data analysts should possess and career opportunities in the field of data analytics.
4. Data Sources and Collection Methods	4. data can come from various internal and external sources, and that appropriate collection methods, such as using public data sets, scripts, and manual entry, are essential for obtaining reliable and relevant information.	8. identify different types of data sources (e.g. internal, external, interview, and real-time data). 9. discuss data collection methods from public datasets scripts and manual data entry.
5. Data Processing	5. effective data processing requires securing data through backups, detecting and addressing missing or inconsistent entries, converting data formats accurately based on given requirements, and refining datasets using tools such as filters and pivot tables to prepare information for deeper analysis.	10. recognize the importance of exploratory data analysis in understanding data and guiding subsequent analysis. 11. create backups of data to maintain security and track changes. 12. identify missing values, duplicates, and inconsistencies in data using spreadsheet functions. 13. convert data formats and types consistently and accurately based on specific requirements. 14. filter pivot table data to refine and focus on relevant subsets for further analysis.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding that...*	LEARNING COMPETENCIES *The learners...*
Data Analysis, Visualization, and Storytelling		
1. Descriptive and Inferential Statistics Using Spreadsheet or Statistical Software	1. descriptive and inferential statistics are essential tools for understanding, summarizing, and analyzing data, and that spreadsheets or statistical software enable them to compute measures, perform tests, and draw meaningful conclusions from datasets.	1. calculate measures of central tendency (mean, median and mode) and variation (range, variance, standard deviation, percentiles and quartiles) using spreadsheet or statistical software. 2. conduct inferential statistical tests (e.g., t-test, z-test, correlation, and regression) using spreadsheet or statistical software.
2. Conditional and Lookup Functions	2. conditional and lookup functions enable efficient data analysis, retrieval, and processing, and that combining these functions supports more complex and dynamic data analysis tasks.	3. apply conditional functions (e.g., AVERAGEIF, AVERAGEIFS, COUNTIF, COUNTIFS, IF, SUMIF, and SUMIFS) to analyze and summarize data based on specific areas. 4. utilize lookup functions (e.g., XLOOKUP, INDEX, MATCH, and VLOOKUP) to retrieve data from different locations within a spreadsheet. 5. construct formulas that combine conditional and lookup functions to perform complex data analysis and processing tasks.
3. Data Visualization Using Spreadsheet	3. data visualization enhances the interpretation and communication of information, and that spreadsheets provide tools such as charts, conditional formatting, tables, pivot tables, and slicers for creating clear and effective visual representations of data.	6. discuss the importance of data visualization. 7. apply conditional formatting, sparklines, and charts (e.g., column, line, pie, bar, scatter plot and combo charts) to visualize data effectively in spreadsheet.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding that...*	LEARNING COMPETENCIES *The learners...*
		8. organize datasets using tables, table slicers, and aggregation methods (e.g., sum, average, and count) to present data. 9. use pivot tables and charts to display multiple aggregation values and perform calculations such as percentages, differences, running totals, and rankings. 10. use report filters, timelines, and slicers to display customized pivot tables and charts.
4. Data Visualization Software	4. data visualization software enables the creation of interactive dashboards by importing data, selecting appropriate visual elements, and applying functions such as sorting, filtering, and drilldown to explore insights dynamically.	11. discuss how to set up a data visualization software environment, import data, and add visual elements (e.g., cards, columns, slicers, and tables) to create interactive dashboards. 12. select appropriate visualizations to represent transformed data effectively. 13. execute data visualization software functions such as editing properties, sorting data, drilldown and filtering, and creating hierarchy
5. Data Storytelling	5. data storytelling is a powerful method of communicating insights by combining visuals, context, and narrative to help audiences understand and act on data.	14. apply effective data storytelling techniques to communicate data insights.
6. Data Analytics Project	6. a data analytics project requires planning, designing, analyzing, and presenting real-world data to apply	15. present a project involving data analytics using real-world data.

CONTENT	CONTENT STANDARDS *The learners demonstrate an understanding that...*	LEARNING COMPETENCIES *The learners…*
	statistical methods, visualization tools, and ethical considerations in solving practical problems.	

Performance Standards

- The learners can illustrate data life cycle and uphold data privacy and ethical principles. They process data accurately and securely by creating backups, cleaning datasets to address missing or inconsistent entries, converting data formats based on specified requirements, and using spreadsheet tools such as filters and pivot tables to refine datasets and generate meaningful insights for decision-making.
- The learners can perform statistical analyses, apply conditional and lookup functions, create visualizations using spreadsheets and data visualization software, communicate insights through data storytelling, and design and present a complete data analytics project using real-world data.